JNTUK R23 REGULATION · SEMESTER I

Basic Civil & Mechanical Engineering

Complete Study Module

UNIT III

Power Plants · Power Transmission · Robotics

Steam Power Plant | Diesel Power Plant | Hydro Power Plant | Nuclear Power Plant

Belt Drives | Chain Drives | Rope Drives | Gear Drives | Robotics

Robot Configurations

Prepared as per JNTUK R23 Syllabus | Exam-Oriented | All Diagrams Included

2-Mark & 5-Mark Questions with Answers | Quick Revision Sheet

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SECTION 01 · PART A

Introduction to Power Plants

Definition: *A power plant (or power station) is an industrial facility that converts one form of energy (chemical, mechanical, nuclear, or potential) into electrical energy for distribution and consumption.*

Need for Power Plants

Electricity is the backbone of modern civilisation. Power plants supply energy to industries, homes, hospitals, transport networks, and communication systems. Without power plants, modern life as we know it would be impossible.

Classification of Power Plants

Type	Energy Source	Fuel / Medium	Scale of Use
Thermal / Steam	Chemical (Coal/Oil/Gas)	Coal, Fuel Oil, Natural Gas	Very Large
Diesel	Chemical (Diesel)	High Speed Diesel	Small to Medium
Hydro	Potential Energy (Water)	Water head	Large
Nuclear	Nuclear Fission	Uranium-235, Plutonium	Very Large
Solar	Solar Radiation	Sunlight	Small to Large
Wind	Kinetic (Wind)	Wind	Medium to Large

General Energy Conversion Chain

Primary Energy → Heat / Mechanical → Turbine / Generator → Electrical Energy → Transmission Line → Consumer / Load

Key Terminology

Term	Meaning
Installed Capacity	Maximum electrical power the plant can produce (in MW or kW)
Load Factor	Ratio of average load to maximum load over a time period

Term	Meaning
Plant Efficiency	η = Electrical Output / Heat Input × 100%
Calorific Value	Heat released by complete combustion of unit mass of fuel (kJ/kg)
Turbine	Machine that converts fluid energy into rotational mechanical energy
Generator	Converts mechanical rotational energy into electrical energy

⚡ BASIC POWER PLANT FORMULAS

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Overall Efficiency η = (Net Electrical Output) / (Heat Input) × 100%
Heat Input (Q) Q = mass of fuel × Calorific Value (kJ)
Power Output P = V × I (Watts) or P = 2πNT / 60 (Watts)
Load Factor LF = Average Load / Maximum Load
Plant Capacity Factor CF = Actual Energy Generated / Maximum Possible
Energy
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SECTION 02

Steam Power Plant (Thermal Power Plant)

Definition: *A Steam Power Plant converts thermal energy from combustion of fossil fuels (coal, oil, gas) into electrical energy by using steam as the working fluid to drive a turbine connected to a generator.*

Major Components

Component	Function
Boiler (Furnace)	Burns fuel; converts water into high-pressure steam
Superheater	Further heats steam to increase temperature without raising pressure
Steam Turbine	Converts steam energy into rotational mechanical energy
Generator (Alternator)	Converts mechanical rotation into electricity
Condenser	Converts exhaust steam back to water
Feed Pump	Pumps condensate back to boiler at high pressure
Economiser	Pre-heats feed water using flue gases
Air Pre-heater	Pre-heats combustion air using flue gases
Cooling Tower	Cools condenser cooling water for reuse

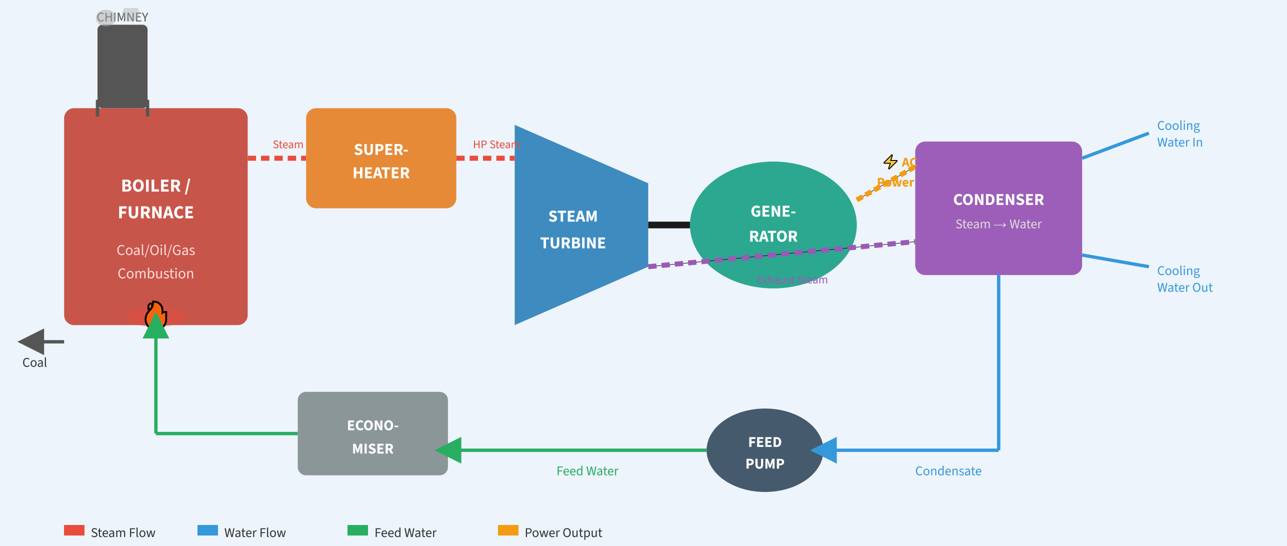
Working Principle (Step-by-Step)

1. **Fuel Combustion:** Coal/oil/gas is fed into the furnace and burned. The heat generated is used to produce steam in the boiler.
2. **Steam Generation:** Water in the boiler absorbs heat and converts into high-pressure, high-temperature steam (typically 150–200 bar, 500–600°C).
3. **Superheating:** Steam passes through the superheater where it is further heated to increase enthalpy and efficiency.
4. **Turbine Rotation:** High-pressure steam expands through the steam turbine blades, imparting kinetic energy and rotating the turbine shaft at high speed (3000 RPM for 50 Hz).
5. **Power Generation:** The turbine shaft is coupled to the generator (alternator) which produces AC electricity.
6. **Condensation:** Exhaust steam (low pressure) from the turbine enters the condenser where it is cooled by water and converts back to liquid water.

7. **Feed Pump & Cycle Repeat:** The condensed water is pumped by the feed pump back into the boiler completing the Rankine Cycle.

8. **Flue Gas Treatment:** Hot flue gases pass through economiser (preheats feed water) and air pre-heater (heats combustion air), before being discharged via chimney.

FIG. 2.1 – STEAM POWER PLANT LAYOUT (SCHEMATIC DIAGRAM)



🔢 STEAM POWER PLANT FORMULAS

Thermal Efficiency $\eta_{th} = W_{net} / Q_{in} = (Q_{in} - Q_{out}) / Q_{in}$

Rankine Efficiency $\eta_R = (h_1 - h_2) / (h_1 - h_{f2})$ [h = enthalpy, kJ/kg]

Steam Consumption $m_s = (P \times 3600) / (h_1 - h_2)$ [kg/kWh]

Heat Rate $HR = 3600 / \eta_{th}$ [kJ/kWh]

Plant Efficiency η = Output (kWh) / [Fuel mass × CV] × 860

✓ Advantages

- High capacity (100–2000 MW)
- Works on multiple fuels (coal, oil, gas)
- Reliable and well-proven technology
- Can be located anywhere
- High load factor (~80%)

✗ Disadvantages

- Lower efficiency (30–40%)
- High greenhouse gas emissions
- Requires large quantity of water
- Slow start-up time (hours)
- High ash & pollution disposal

🏭 Applications

- Base-load electricity generation for cities & industrial zones
- Grid power supply in India (e.g., NTPC plants, Vijayawada TPS)
- Cogeneration (combined heat & power) in industries

SECTION 03

Diesel Power Plant

Definition: *A Diesel Power Plant uses a diesel engine (internal combustion engine operating on the diesel cycle) as a prime mover to drive a generator for producing electrical energy. It works on the principle of fuel ignition by high compression of air.*

Major Components

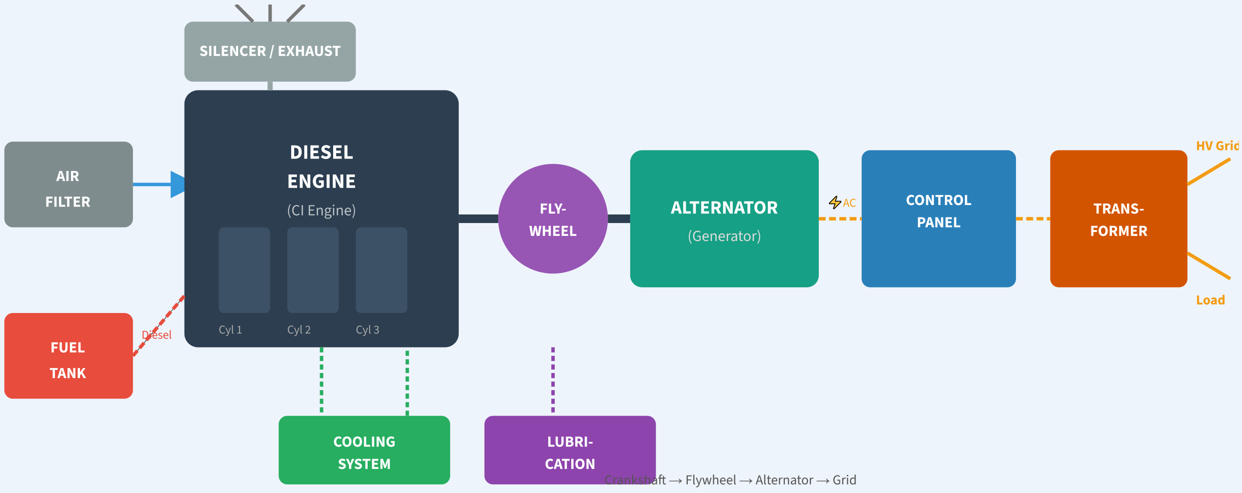
Component	Function
Diesel Engine (CI Engine)	Burns diesel fuel to produce mechanical power (prime mover)
Air Intake System	Filters and supplies fresh air for combustion
Fuel Injection System	Injects diesel at high pressure into cylinders
Exhaust System	Removes combustion gases via silencer & exhaust pipe
Cooling System	Maintains engine temperature (water/air cooling)
Lubrication System	Reduces friction between moving parts
Generator (Alternator)	Coupled to engine shaft; produces AC electricity
Starting System	Electric motor or compressed air to start the engine
Control Panel	Monitors and controls electrical output, protection relays

Working Principle – Diesel Cycle (Step-by-Step)

1. **Suction Stroke:** The piston moves downward (BDC), drawing in fresh air through the intake valve. Inlet valve is open; exhaust valve is closed.
2. **Compression Stroke:** Both valves close. Piston moves upward (TDC), compressing air to a compression ratio of 14:1 to 22:1. Air temperature rises to ~550–700°C.
3. **Power (Expansion) Stroke:** Diesel is injected into the hot compressed air at high pressure. It ignites spontaneously (no spark plug). Combustion raises pressure and pushes piston downward, producing power.
4. **Exhaust Stroke:** Exhaust valve opens, piston moves upward pushing out burnt gases through the exhaust system (via silencer to atmosphere).
5. **Power Transmission:** The reciprocating motion of pistons is converted to rotational motion by the crankshaft, which drives the alternator coupled to it via a flywheel.

6 **Electricity Generation:** The rotating alternator produces 3-phase AC electricity which is stepped up by transformer and fed to the grid or local loads.

FIG. 3.1 – DIESEL POWER PLANT LAYOUT (SCHEMATIC)



🔢 DIESEL ENGINE / POWER PLANT FORMULAS

Compression Ratio $r = V_{BDC} / V_{TDC}$ (typically 14 to 22)

Cut-off Ratio $r_c = V_3 / V_2$ (ratio of volumes at fuel cut-off)

Diesel Cycle Eff. $\eta_D = 1 - (1/r^{(\gamma-1)}) \times [(r_c^\gamma - 1) / \gamma(r_c - 1)]$

Indicated Power (IP) $IP = P_m \times L \times A \times N / 60$ [Watts]

Brake Power (BP) $BP = 2\pi NT / 60$ [Watts]

Mechanical Eff. $\eta_m = BP / IP \times 100\%$

Heat Rate HR = Fuel consumed (kJ) / Power output (kWh)

✓ Advantages

- Quick starting (within minutes)
- High efficiency at part load
- No water requirement for steam
- Compact and portable
- Suitable for standby/emergency power

✗ Disadvantages

- High fuel cost (diesel is expensive)
- Limited capacity (up to ~50 MW)
- High maintenance cost
- Noise & vibration issues
- Not suitable for base load

🏭 Applications

- Standby/emergency power for hospitals, data centers, banks
- Remote area & island electrification
- Peak load shaving in power systems
- Mobile power stations for military & construction

SECTION 04

Hydro Power Plant (Hydroelectric Power Station)

Definition: *A Hydro Power Plant converts the potential energy of stored water at a height into kinetic energy and then into electrical energy using water turbines coupled to generators.*

Major Components

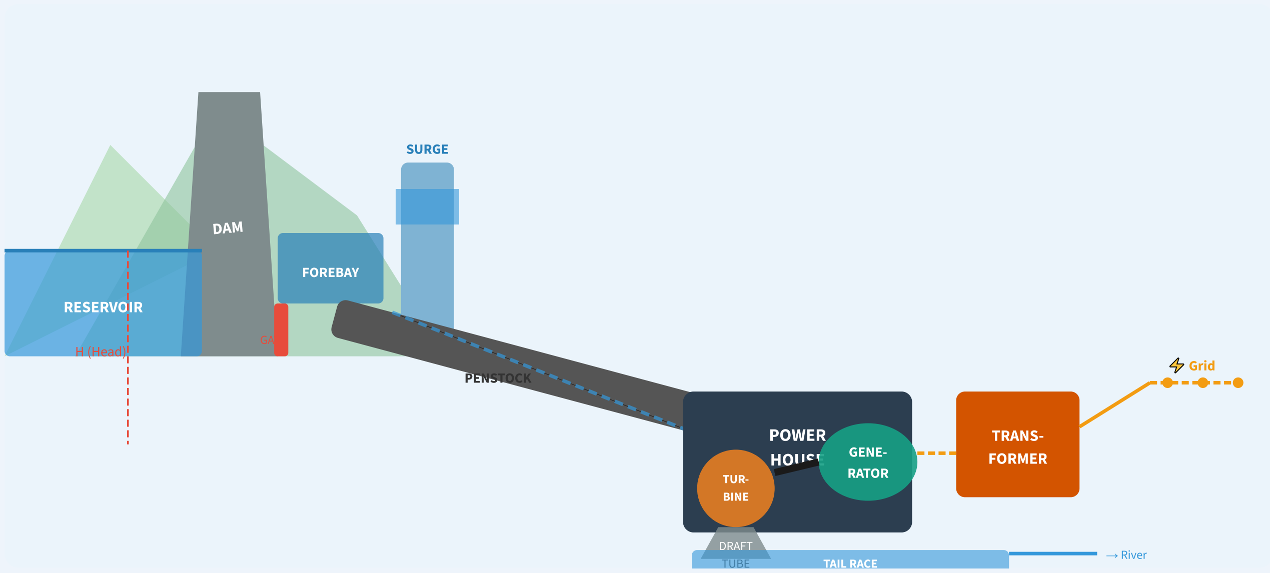
Component	Function
Dam / Reservoir	Stores water at height to create water head (potential energy)
Forebay (Head Pond)	Stores water just before the penstock; regulates flow
Penstock	Large pipes carrying water from reservoir to turbine at high velocity
Surge Tank	Absorbs sudden water pressure changes (water hammer effect)
Water Turbine	Converts kinetic/pressure energy of water into rotation (Pelton/Francis/Kaplan)
Generator / Alternator	Coupled to turbine; converts rotation to electricity
Draft Tube	Discharges water from turbine exit to tail race smoothly
Tail Race	Lower water channel carrying discharged water back to river
Control Gates / Valves	Control water flow; used for shutdown and regulation

Working Principle (Step-by-Step)

1. **Water Storage:** Rainwater is stored in a large reservoir behind a dam. The height of water above the turbine is called the *effective head (H)*.
2. **Water Flow to Penstock:** Water is released from the reservoir through gates and flows through the penstock (large diameter pipe) at increasing velocity due to gravity.
3. **Surge Tank Function:** When load suddenly decreases, the surge tank absorbs excess water pressure, preventing pipe damage (water hammer).
4. **Turbine Rotation:** High-velocity water strikes the turbine blades (Pelton/Francis/Kaplan depending on head and flow), rotating the turbine shaft at high speed.
5. **Power Generation:** The turbine shaft drives the generator/alternator which produces 3-phase AC electricity.
6. **Draft Tube & Tail Race:** Water exits through the draft tube (which converts kinetic energy to pressure, improving efficiency) into the tail race and returns to the river.

7 **Step-Up Transformer:** Generated electricity is stepped up by transformer for long-distance transmission via grid.

FIG. 4.1 – HYDRO POWER PLANT SCHEMATIC DIAGRAM



🔢 HYDRO POWER PLANT FORMULAS

Potential Energy PE = mgh [Joules] (m=mass in kg, $g=9.81\ m/s^2$, h=head in m)

Power Available $P = \rho \times g \times Q \times H \times \eta$ [Watts]

($\rho = 1000\ kg/m^3$, Q = flow rate m^3/s, H = head m)

Hydraulic Efficiency η_h = Power at turbine / Water Power

Overall Efficiency η = Electrical Output / ($\rho g Q H$) × 100%

Specific Speed $N_s = N \times \sqrt{P} / H^{(5/4)}$ (turbine selection parameter)

✓ Advantages

- Renewable & clean energy (zero fuel cost)
- Very high efficiency (85–95%)
- Long life span (50–100 years)
- No air pollution or CO_2
- Quick start and load following

✗ Disadvantages

- Dependent on rainfall & geography
- High initial capital cost
- Ecological impact & displacement
- Silting of reservoir over time
- Long gestation period (10–15 years)

🏭 Applications

- Large-scale power generation (e.g., Nagarjunasagar, Srisailam in AP)
- Peak load shaving (pump storage plants)
- Irrigation water supply (multi-purpose dams)
- Flood control & drinking water storage

SECTION 05

Nuclear Power Plant

Definition: *A Nuclear Power Plant uses the heat generated by nuclear fission (splitting of heavy atomic nuclei like U-235) to produce steam, which drives a turbine-generator system to generate electricity.*

Major Components

Component	Function
Nuclear Reactor	Core where nuclear fission occurs; houses fuel rods, moderator, and control rods
Fuel Rods	Contains enriched U-235 fuel pellets; fission occurs here
Control Rods (Boron/Cadmium)	Absorb neutrons to control reaction rate; inserted/withdrawn to increase/decrease power
Moderator (Water/Heavy Water)	Slows down neutrons to thermal speeds for sustained chain reaction
Coolant (Primary Circuit)	Removes heat from reactor core (pressurized water, heavy water)
Heat Exchanger / Steam Generator	Transfers heat from primary coolant to secondary water circuit
Steam Turbine	Steam drives turbine to produce mechanical energy
Generator	Converts mechanical energy into electrical energy
Condenser & Cooling Tower	Condenses exhaust steam back to water
Biological Shield	Thick concrete/lead walls prevent radiation leakage

Working Principle (Step-by-Step)

1. **Nuclear Fission:** Uranium-235 nuclei absorb slow (thermal) neutrons and split into smaller nuclei (krypton + barium), releasing enormous energy (≈200 MeV/fission) and 2–3 new neutrons.
2. **Chain Reaction:** Released neutrons cause further fissions — a controlled, self-sustaining chain reaction occurs in the reactor core.
3. **Control Rod Regulation:** Boron or cadmium control rods absorb excess neutrons to maintain criticality (controlled fission rate).

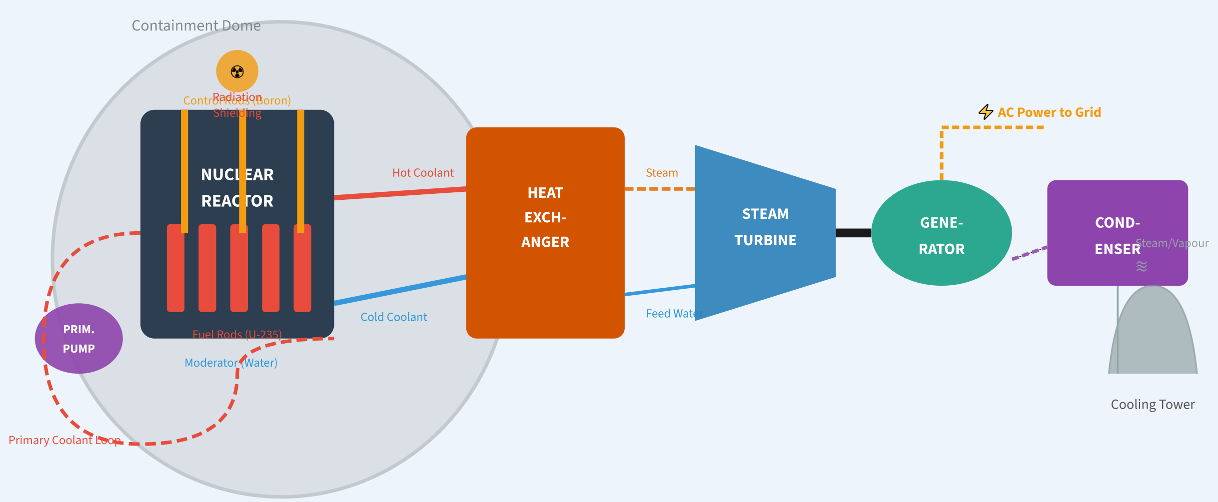
4 **Heat Removal (Primary Loop):** Coolant (pressurized water) circulates around fuel rods, absorbing heat (typically 300–330°C) without boiling due to high pressure.

5 **Steam Generation (Secondary Loop):** In the heat exchanger/steam generator, primary coolant heats secondary water to generate steam — the two loops are separate (no radiation in secondary circuit).

6 **Turbine-Generator:** Steam drives the turbine which is coupled to generator, producing electricity.

7 **Condenser & Safety:** Steam condenses; cooling towers reject waste heat. Biological shield and containment dome prevent radiation release.

FIG. 5.1 – NUCLEAR POWER PLANT SCHEMATIC



NUCLEAR PHYSICS & PLANT FORMULAS

Mass-Energy Equivalence $E = mc^2$ ($c = 3\times10^8$ m/s)

Energy per fission ≈ 200 MeV = 3.2×10^{-11} J (for U-235)

Fissions per second = Power (W) / ($200\times10^6 \times 1.6\times10^{-19}$) fissions/s

Thermal Efficiency $\eta \approx$ 33–40% (similar to steam plant, Rankine cycle)

Enrichment Natural U: 0.7% U-235; Enriched: 3–5% U-235

✓ Advantages

- Huge energy from small fuel mass
- No CO_2 / greenhouse gas emissions
- Very high capacity (1000–3000 MW)
- Low fuel cost per unit energy
- Reliable base-load power

✗ Disadvantages

- Very high capital & construction cost
- Radioactive waste disposal problem
- Risk of meltdown & radiation
- Long construction time (10–20 years)
- High decommissioning cost

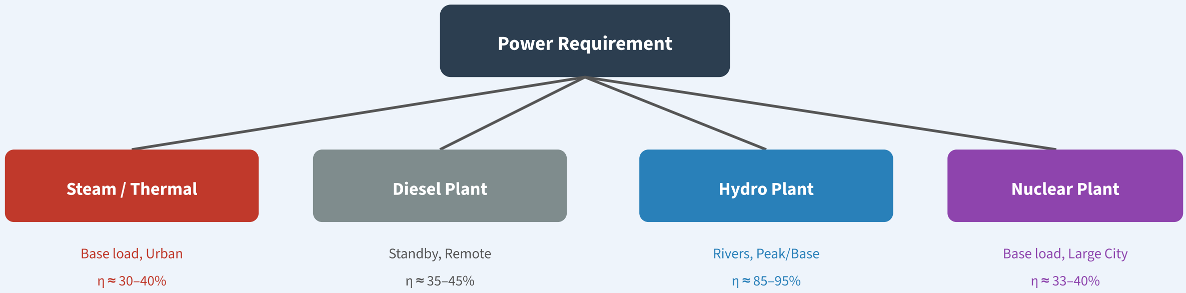
SECTION 06

Comparison of Power Plants

Parameter	Steam (Thermal)	Diesel	Hydro	Nuclear
Energy Source	Coal / Oil / Gas	Diesel	Water Head	U-235 / Pu-239
Working Fluid	Steam (water)	Air-fuel mixture	Water	Steam (water)
Prime Mover	Steam Turbine	CI Engine	Water Turbine	Steam Turbine
Efficiency	30–40%	35–45%	85–95%	33–40%
Typical Capacity	100–2000 MW	1–50 MW	5–10000 MW	500–3000 MW
Capital Cost	Medium	Low	Very High	Very High
Running Cost	Medium (fuel)	Very High	Very Low	Low
Start-Up Time	4–8 hours	Minutes	Seconds/Min	Hours/Days
Location	Anywhere	Anywhere	Near rivers	Remote areas
Environmental Impact	High (CO_2, ash)	High (exhaust)	Low	Low (radiation risk)
Renewable	No	No	Yes	No
Use Type	Base load	Standby / Peak	Base / Peak	Base load
Water Requirement	Very High	Low	Very High (natural)	Very High
Skilled Personnel	Medium	Low	Low	Very High
Indian Examples	NTPC Vijayawada TPS	Remote villages	Nagarjunasagar, Srisailam	Kudankulam, Tarapur

Summary Flowchart

FIG. 6.1 – POWER PLANT SELECTION GUIDE



SECTION 07 · PART B

Mechanical Power Transmission – Introduction

Definition: *Mechanical Power Transmission is the transfer of mechanical energy (torque and speed) from a power source (prime mover/motor) to a driven machine using mechanical elements such as belts, chains, ropes, or gears.*

Why is Power Transmission Needed?

In most engineering applications, the power source (motor/engine) and the machine to be driven are separated by a distance or their speeds are different. Power transmission systems bridge this gap by transmitting power efficiently while allowing speed change, direction change, and torque multiplication.

Types of Mechanical Power Transmission

Type	Drive Element	Connection	Speed Ratio	Slip
Belt Drive	Flat/V/Ribbed Belt	Flexible (friction)	Variable (slip exists)	Yes (1–2%)
Chain Drive	Roller/Silent Chain	Positive (mesh)	Exact (no slip)	No
Rope Drive	Hemp/Wire Rope	Flexible (friction)	Variable (slip exists)	Yes
Gear Drive	Spur/Helical/Bevel Gears	Positive (mesh)	Exact (no slip)	No

Key Parameters in Power Transmission

🔢 **GENERAL POWER TRANSMISSION FORMULAS**

Power (P) $P = 2\pi NT / 60$ [Watts] (N = RPM, T = Torque in N·m)

Power (P) $P = F \times v$ [Watts] (F = Force in N, v = velocity m/s)

Velocity Ratio (VR) $VR = N_1 / N_2 = D_2 / D_1$ (for belt/rope drives)

Speed Ratio (SR) SR = Driver speed / Driven speed = T_{driven} / T_{driver} (gears)

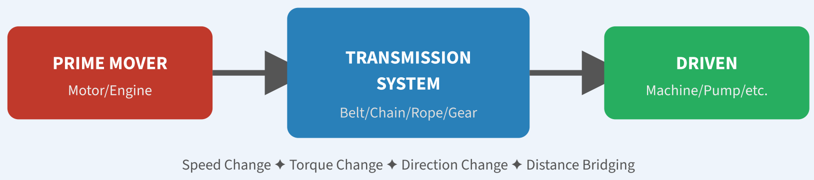
Efficiency η = Power Output / Power Input × 100%

Belt Velocity $v = \pi \times D \times N / 60$ [m/s]

Torque $T = P \times 60 / (2\pi N)$ [N·m]

Transmission Angle $\theta = 180° \pm 2\alpha$ (angle of wrap on pulleys)

FIG. 7.1 – POWER TRANSMISSION SYSTEM OVERVIEW



SECTION 08

Belt Drives

Definition: *A Belt Drive is a power transmission system that uses a flexible belt (looped over two or more pulleys) to transmit power from one shaft to another by means of* ***friction*** *between the belt and pulley surfaces.*

Types of Belt Drives

Type	Belt Cross-Section	Feature	Application
Flat Belt Drive	Rectangular	Long distance, light load	Textile mills, conveyors
V-Belt Drive	Trapezoidal (wedge)	High power, compact	Automotive, machine tools
Ribbed (Poly-V) Belt	Multi-ribbed flat	High speed, thin section	Automotive accessories
Timing Belt (Toothed)	Toothed profile	No slip (positive drive)	Camshafts, CNC machines
Circular Belt (Round)	Circular	Grooved pulleys, low power	Sewing machines, watches

Types of Belt Drive Arrangements

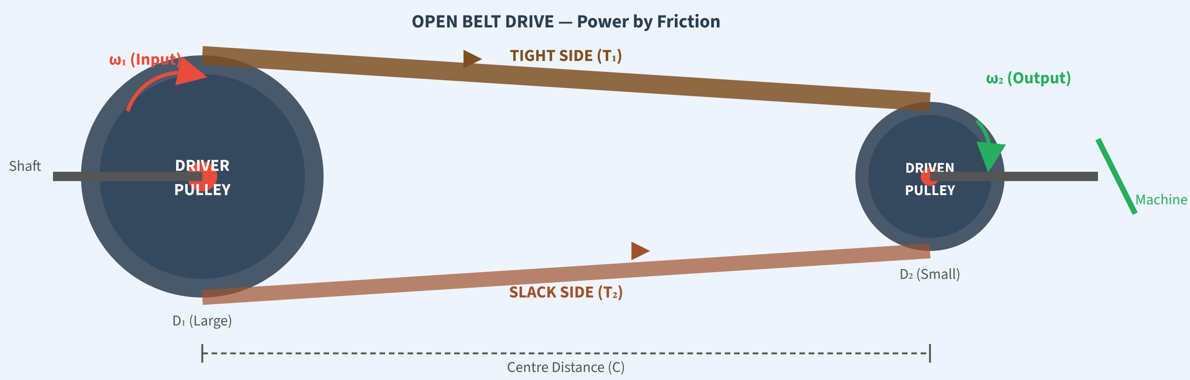
Open Belt Drive

- Both pulleys rotate in the SAME direction
- Suitable when driver and driven are on the same side
- Angle of wrap = $180° - 2\alpha$
- Used in most industrial applications

Crossed (Twist) Belt Drive

- Pulleys rotate in OPPOSITE directions
- Belt twisted between pulleys (8-shaped path)
- Angle of wrap = $180° + 2\alpha$
- Greater power capacity; belt wears faster

FIG. 8.1 – BELT DRIVE SYSTEM (OPEN BELT DRIVE)



🔢 BELT DRIVE FORMULAS

Velocity Ratio $VR = N_1/N_2 = D_2/D_1$ (ignoring slip)

With Slip $VR = D_2(1-s_1/100) / D_1(1-s_2/100) \approx D_2/D_1 \times (1 - s/100)$

Belt Velocity $v = \pi D_1 N_1 / 60 = \pi D_2 N_2 / 60$ [m/s]

Power Transmitted $P = (T_1 - T_2) \times v$ [Watts]

Ratio of Belt Tensions $T_1/T_2 = e^{(\mu\theta)}$ (μ = friction coeff, θ = angle of wrap in rad)

Length of Open Belt $L = \pi(D_1+D_2)/2 + 2C + (D_1-D_2)^2/4C$

Angle of Wrap $\theta = 180° - 2 \sin^{-1}[(D_1-D_2)/2C]$ (open belt)

Centrifugal Tension $Tc = mv^2$ (m = mass per unit length, v = belt speed)

✓ Advantages

- Smooth & quiet operation
- Acts as safety (slips on overload)
- Low cost, easy maintenance
- Can transmit over long distances
- Shock absorbing (flexible)

✗ Disadvantages

- Slip causes non-exact speed ratio
- Not suitable for exact timing
- Belt deteriorates with heat/oil
- Speed limited (generally <25 m/s)
- Higher shaft loads due to pre-tension

SECTION 09

Chain Drives & Rope Drives

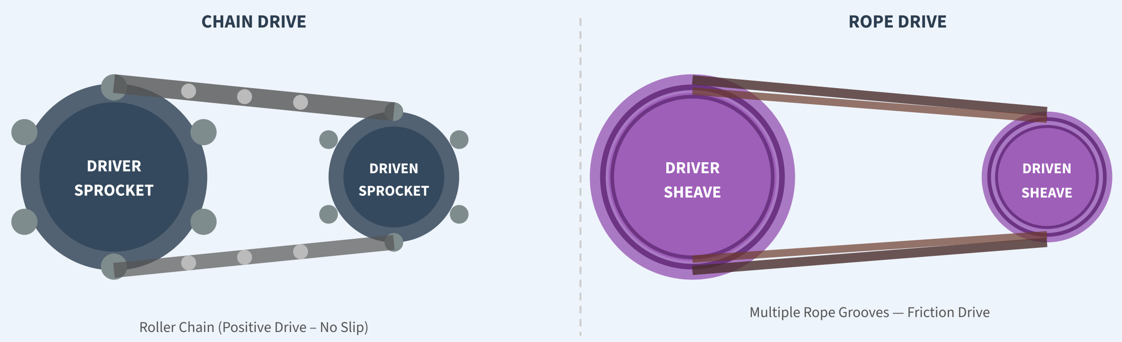
9.1 Chain Drives

> ***Definition:*** *A Chain Drive transmits power through a chain that meshes with toothed sprockets on driver and driven shafts. It is a positive (non-slip) drive.*

Types of Chains

Type	Feature	Application
Roller Chain (most common)	Rollers reduce friction; standard ANSI sizes	Bicycles, motorcycles, conveyors
Silent (Inverted Tooth) Chain	Smooth, quiet, high speed	Machine tools, automotive camshaft
Leaf Chain	High tensile strength, no sprocket	Forklift masts, balancing
Flat-Link Chain	Simple, low speed	Conveyors, hoists

FIG. 9.1 – CHAIN DRIVE & ROPE DRIVE SCHEMATICS



CHAIN DRIVE FORMULAS

Velocity Ratio $VR = N_1 / N_2 = T_2 / T_1$ (T = number of teeth on sprocket)

Chain Speed $v = (T \times p \times N) / 60$ [m/s] (p = pitch in m, N = RPM)

Power $P = (T_1 - T_2) \times v$ [Watts]

9.2 Rope Drives

> ***Definition:*** *A Rope Drive uses ropes (natural fibre or wire) running in grooved pulleys (sheaves) to transmit power over long distances through friction. Multiple ropes can be used to increase power capacity.*

Types of Ropes

Type	Material	Application
Fibre / Hemp Rope	Natural fibre (manila, hemp)	Hoists, lifting, light drives
Wire Rope	High-tensile steel wires (6×7, 6×19, 6×37)	Mine hoists, elevators, cranes, bridges

✓ Chain Drive Advantages

- No slip — exact velocity ratio
- High efficiency (~98%)
- Suitable for short & medium distances
- Works in oily/wet conditions
- Can drive multiple sprockets

✗ Chain Drive Disadvantages

- Noisy at high speeds
- Requires periodic lubrication
- Polygonal effect causes vibration
- Limited to parallel shafts
- Stretches over time

🏭 Applications of Chain Drives

- Bicycles, motorcycles, and automobile timing systems
- Agricultural machines (threshers, tractors)
- Conveyor systems in manufacturing
- Hoist and crane mechanisms

🏭 Applications of Rope Drives

- Mine shaft hoists and winding drums
- Elevator (lift) suspension cables
- Bridge cables and suspension systems
- Cranes, derricks, pulley blocks

SECTION 10

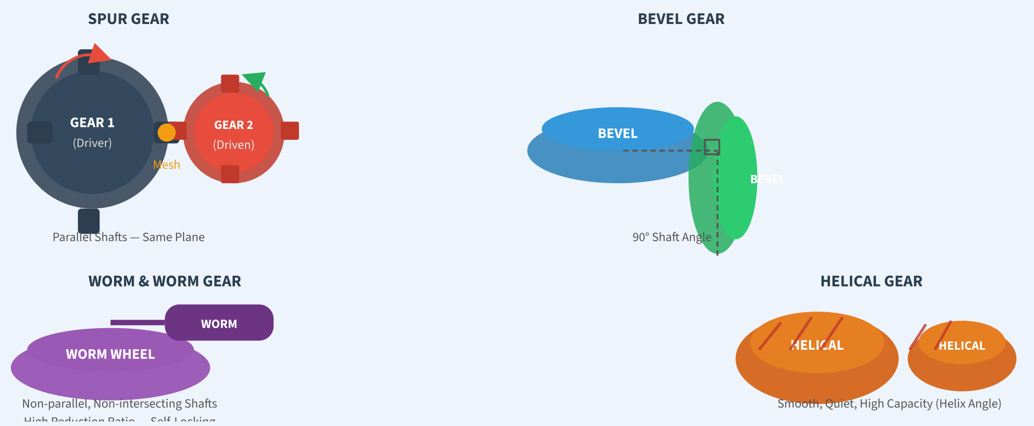
Gear Drives

Definition: *A Gear Drive transmits power between shafts by means of* ***toothed wheels (gears)*** *that mesh together. The meshing teeth provide a positive (non-slip) drive with a definite velocity ratio.*

Classification of Gears

Shaft Orientation	Gear Type	Characteristics	Applications
Parallel Shafts	Spur Gear	Straight teeth, noisy, common	Clocks, simple gearboxes
Parallel Shafts	Helical Gear	Inclined teeth, smooth, quiet, high load	Automotive transmission
Parallel Shafts	Rack & Pinion	Rotary to linear motion conversion	Steering, CNC machines
Intersecting Shafts	Bevel Gear (Straight)	Conical shape, 90° shaft angle	Differential, hand drill
Intersecting Shafts	Spiral Bevel Gear	Curved teeth, smooth operation	Automotive rear axle
Non-parallel, Non-intersecting	Worm & Worm Gear	High reduction ratio, self-locking	Lifts, cranes, reduction boxes
Non-parallel, Non-intersecting	Hypoid Gear	Offset axes, very smooth	Automotive drive axle

FIG. 10.1 – GEAR DRIVE TYPES (SPUR, HELICAL, BEVEL, WORM)



🔢 GEAR DRIVE FORMULAS

Gear Ratio (i) $i = N_1 / N_2 = T_2 / T_1$ (N = speed, T = teeth count)

Module (m) $m = D / T = p / \pi$ (D = pitch circle diameter, p = circular pitch)

Pitch Circle Dia. $D = m \times T$

Centre Distance $C = m(T_1 + T_2) / 2$

Train Value (T.V.) TV = Product of driving teeth / Product of driven teeth

Speed of Last Gear $N_{last} = N_{first} \times TV$

Worm Gear Ratio VR = $T_{worm\ wheel}$ / (No. of starts on worm)

Efficiency of Worm $\eta = \tan(\lambda) / \tan(\lambda + \phi)$ (λ = lead angle, ϕ = friction angle)

✓ Advantages of Gear Drives

- No slip — exact speed ratio
- Very high efficiency (96–99%)
- Compact and robust
- Long life (metal gears)
- Can transmit high power
- Multiple gear ratios (gearbox)

✗ Disadvantages of Gear Drives

- High manufacturing cost
- Noise at high speeds
- Require lubrication
- No overload protection (rigid)
- Error in tooth profile causes vibration

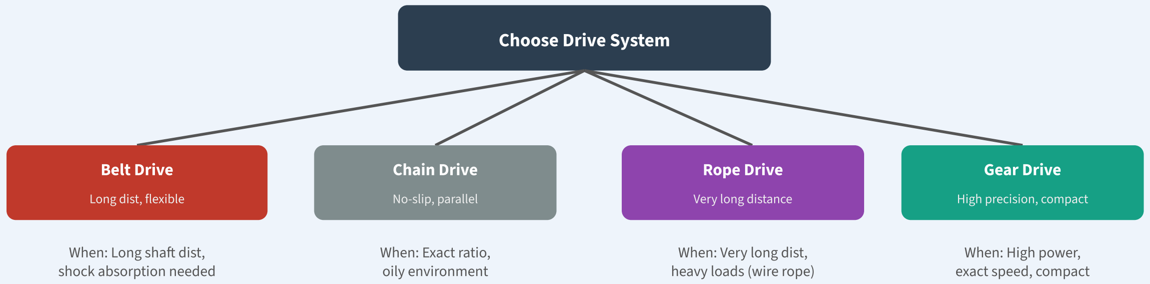
SECTION 11

Comparison of Power Transmission Systems

Parameter	Belt Drive	Chain Drive	Rope Drive	Gear Drive
Type of Drive	Friction	Positive (mesh)	Friction	Positive (mesh)
Slip	Yes (1–2%)	No	Yes	No
Speed Ratio	Variable (slip)	Constant	Variable	Constant
Efficiency	95–98%	97–99%	90–95%	96–99%
Velocity Range	5–25 m/s (flat); up to 50 m/s (V)	0.5–15 m/s	10–60 m/s	Up to 200 m/s (pitch line)
Power Range	Low to medium	Medium	High (wire)	Very High
Distance	Long (up to 15 m)	Short–medium (up to 3 m)	Very Long	Very Short (adjacent)
Lubrication	Not required	Required	Not required	Required
Noise	Silent	Noisy at high speed	Silent	Noisy (spur); quiet (helical)
Overload Protection	Yes (slips)	No	Yes (slips)	No
Maintenance	Easy (replace belt)	Moderate (lubricate)	Easy	Difficult (precision)
Cost	Low	Medium	Low–Medium	High
Shock Absorption	Good (flexible)	Moderate	Good	Poor (rigid)
Shaft Orientation	Parallel / crossed	Parallel only	Parallel	Any angle (type dependent)
Examples	Lathe, textile mill, fan	Bicycle, motorcycle, conveyor	Mine hoist, crane, elevator	Gearbox, differential, clock

Power Transmission Selection Flowchart

FIG. 11.1 – DRIVE SELECTION LOGIC



SECTION 12 · PART C

Introduction to Robotics

Definition (Robot – ISO 8373): *A Robot is an automatically controlled, reprogrammable, multipurpose manipulator that is programmable in three or more axes, which may be either fixed in place or mobile for use in automation applications.*

Robotics *is the interdisciplinary branch of science & engineering dealing with the design, construction, operation, and application of robots.*

Components of a Robot

Component	Function
Manipulator (Robot Arm)	Physical structure of links and joints that performs motion
End Effector	Tool at the end of arm (gripper, welding torch, drill, camera)
Actuators	Drive joints — motors (electric, hydraulic, pneumatic)
Sensors	Detect environment — position, force, vision, proximity sensors
Controller	Computer brain — processes sensor data & controls actuators
Power Supply	Electric, hydraulic, or pneumatic power source
Teach Pendant	Hand-held device to program/teach the robot positions

Joints & Links

A robot arm consists of rigid bodies called **links** connected by **joints**. The joint allows relative motion between adjacent links.

Types of Joints

Joint Type	Symbol	Motion	Degrees of Freedom	Example
Revolute (Rotary)	R	Rotation about an axis	1 (rotational)	Elbow, shoulder rotation
Prismatic (Sliding)	P	Linear translation along axis	1 (translational)	Telescopic arm, cylinder piston

Joint Type	Symbol	Motion	Degrees of Freedom	Example
Cylindrical	C	Rotation + Translation about same axis	2	Rotary-sliding motion
Spherical (Ball)	S	Rotation in 3 axes (ball & socket)	3	Human shoulder joint
Screw (Helical)	H	Rotation + Translation coupled	1 (coupled)	Nut & bolt, lead screw
Planar	Pl	Translation in a plane	2 (translational)	XY table

Degrees of Freedom (DOF)

> ***Degree of Freedom (DOF):*** *The number of independent motions (or parameters) required to completely describe the position and orientation of a robot's end effector. Most industrial robots have 6 DOF (3 for position + 3 for orientation).*

🔢 **ROBOTICS FORMULAS**

DOF of robot = Sum of DOF of all individual joints

Grübler's Formula $F = 3(n-1) - 2j_1 - j_2$ (planar mechanism; n=links, j_1=lower pair joints, j_2=higher pair)

For 3D (spatial): $F = 6(n-1) - 5j_1 - 4j_2 - 3j_3 - 2j_4 - j_5$

Workspace Volume Depends on link lengths (L) and joint limits (θ_{min} to θ_{max})

Payload Maximum mass the robot can handle at end effector (kg)

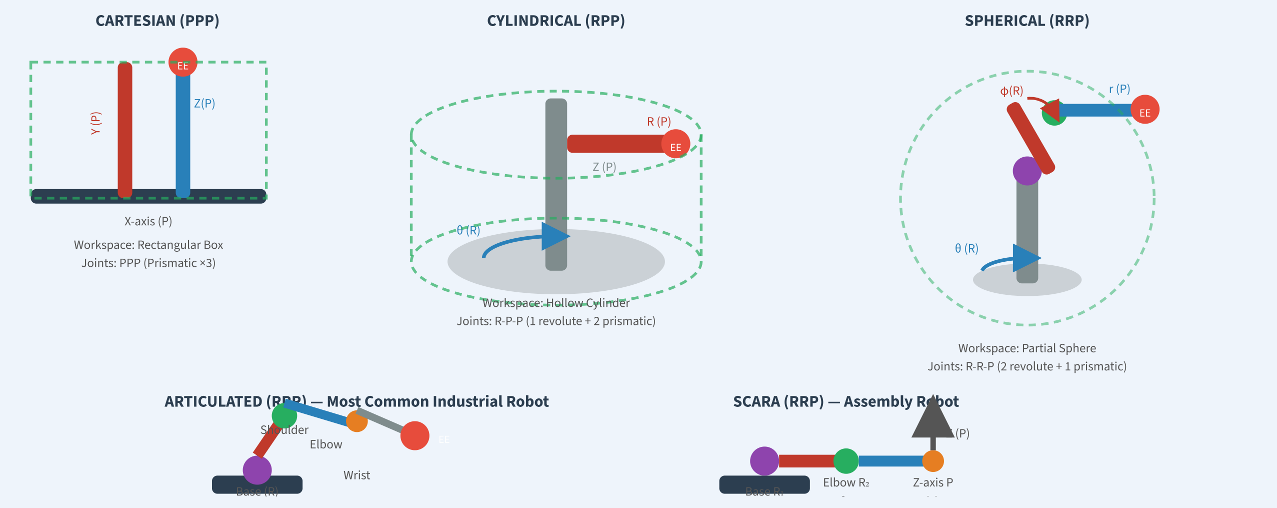
Repeatability Ability to return to same position (±0.01 mm for precision robots)

Robot Configurations & Applications

The **robot configuration** (or robot geometry) defines the arrangement of joints and links that determines the shape of the robot's workspace (work envelope).

Major Robot Configurations

FIG. 13.1 – ROBOT ARM CONFIGURATIONS



Robot Configuration Comparison Table

Configuration	Joint Notation	Workspace Shape	DOF	Advantage	Application
Cartesian	PPP	Rectangular Box	3	High precision, simple programming	CNC machines, gantry robots, 3D printers
Cylindrical	RPP	Hollow Cylinder	3	Large horizontal reach	Machine loading, spot welding
Spherical (Polar)	RRP	Partial Sphere	3	Large work volume	Die casting, forging, material handling
Articulated (Revolute)	RRR	Irregular (large)	6	Most flexible, human-arm-like	Arc welding, spray painting, assembly
SCARA	RRP	Cylindrical (flat)	3–4	Fast, high repeatability	PCB assembly, electronic pick & place

Configuration	Joint Notation	Workspace Shape	DOF	Advantage	Application
Parallel (Delta)	RRR (parallel)	Limited hemisphere	3–6	Very high speed and rigidity	Food packaging, pharmaceutical, sorting

Applications of Robotics

Industry	Robot Application
Automotive	Spot welding, arc welding, spray painting, assembly, inspection
Electronics / PCB	SMT pick & place, soldering, inspection (AOI), IC packaging
Healthcare / Medical	Surgical robots (Da Vinci), drug dispensing, rehabilitation
Food Processing	Packaging, sorting, palletising, cutting, dispensing
Aerospace	Drilling, riveting, composite layup, inspection of aircraft parts
Military & Space	Bomb disposal, exploration rovers (Mars Rover), surveillance drones
Agriculture	Crop harvesting, spraying, seeding, greenhouse monitoring
Logistics & Warehouse	Automated guided vehicles (AGVs), order picking (Amazon Kiva)
Construction	Bricklaying robots, 3D concrete printing, demolition
Education & Research	Humanoid robots (ASIMO, Sophia), teaching aids, AI research

SECTION 14

2-Mark Questions & Answers

2 Marks Q1. Define a power plant. Name its four main types.

Definition: A power plant is an industrial installation where energy in one form (chemical, nuclear, hydraulic) is converted into electrical energy using prime movers like turbines or engines coupled to generators.

Four Main Types:

1. **Steam (Thermal) Power Plant** – uses coal/oil/gas combustion to generate steam for turbine.

2. **Diesel Power Plant** – uses diesel CI engine as prime mover.

3. **Hydro Power Plant** – uses potential energy of water head through water turbines.

4. **Nuclear Power Plant** – uses nuclear fission of uranium to generate heat & steam.

2 Marks Q2. What is the working principle of a steam power plant?

A steam power plant works on the **Rankine Cycle**. Coal is burned in the furnace to produce heat, which converts water in the boiler into high-pressure steam. This steam expands through the steam turbine, rotating it and driving the coupled generator to produce electricity. The exhaust steam is condensed back to water in the condenser, and the feed pump circulates it back to the boiler — completing the closed cycle.

Key Conversion: Chemical Energy → Heat Energy → Mechanical Energy → Electrical Energy

2 Marks Q3. What is a Penstock? State its function in a hydro power plant.

A **penstock** is a large-diameter, thick-walled pipe (usually made of steel or concrete) that carries water from the reservoir (or forebay) to the water turbine in the powerhouse.

Functions:

1. Conveys water from the high-level reservoir to the low-level turbine, converting potential energy to kinetic/pressure energy during flow.

2. Must withstand high water pressure and handle water hammer effects (sudden pressure surges when flow is shut off).

3. The penstock includes isolating valves to shut off water flow for maintenance.

2 Marks **Q4. Differentiate between open belt drive and crossed belt drive.**

Open Belt Drive: Both pulleys rotate in the *same direction*. The belt runs straight between the pulleys without twisting. Angle of wrap = 180° – 2α (α = angle based on pulley size difference). Used when both shafts rotate in the same direction is required.

Crossed Belt Drive: The pulleys rotate in *opposite directions*. The belt crosses itself (forms a figure-8) between the pulleys. Angle of wrap = 180° + 2α, so it has a higher power-transmitting capacity, but the belt wears out faster at the cross-point. Used when reverse rotation of driven shaft is required.

2 Marks **Q5. What is the velocity ratio in belt drives? Give its formula.**

Velocity Ratio (VR) is the ratio of the speed of the driven pulley to the speed of the driver pulley. Equivalently, it is the ratio of the driver pulley diameter to the driven pulley diameter.

Formula:

$VR = N_2 / N_1 = D_1 / D_2$ (ignoring slip)

With slip: $VR = D_1 / D_2 \times (1 - s/100)$ where s = percentage slip

Where: N_1 = Driver speed (RPM), N_2 = Driven speed (RPM), D_1 = Driver diameter, D_2 = Driven diameter.

2 Marks **Q6. Define a gear. State the law of gearing.**

Gear: A gear is a toothed mechanical element (wheel or cylinder) used to transmit power and motion between shafts through the meshing of teeth. Gears provide a definite velocity ratio without slip.

Law of Gearing: The common normal at the point of contact between two meshing gears must always pass through a fixed point on the line joining the centres of the two gears. This fixed point is called the *pitch point* (P). This law ensures a constant velocity ratio between the gears throughout the meshing cycle.

Mathematical statement: Velocity ratio = $N_1/N_2 = T_2/T_1$ = constant throughout meshing

2 Marks **Q7. Define a robot. What are degrees of freedom (DOF)?**

Robot (ISO 8373): An automatically controlled, reprogrammable, multipurpose manipulator programmable in three or more axes, used in industrial automation applications.

Degrees of Freedom (DOF): DOF is the number of independent motions (movements) that a robot's end effector can perform. Each joint provides one or more DOF. A robot with 6 DOF (3 for position + 3 for orientation) can reach any point in its workspace in any orientation. More DOF = more flexibility but more complex control.

2 Marks **Q8. What is nuclear fission? Give the reaction equation.**

Nuclear Fission: Nuclear fission is the process in which a heavy atomic nucleus (like U-235) absorbs a slow (thermal) neutron and splits into two smaller nuclei (fission fragments), releasing a large amount of energy (≈200 MeV per fission) and 2–3 additional neutrons. These released neutrons can trigger further fissions, leading to a *chain reaction*.

Reaction: $^{235}U + {}^{1}n \longrightarrow {}^{92}Kr + {}^{141}Ba + 3\,{}^{1}n + \text{Energy}$ (≈200 MeV)

Energy from mass-energy equivalence: $E = mc^2$ $(c = 3 \times 10^8 \text{ m/s})$

SECTION 15

5-Mark Questions & Answers

5 Marks Q1. Explain the working principle of a Steam Power Plant with a neat diagram.

Introduction: A Steam (Thermal) Power Plant converts chemical energy of fossil fuels into electrical energy using steam as the working fluid, following the Rankine Cycle.

Main Components: Boiler (furnace), superheater, steam turbine, generator, condenser, feed pump, economiser, and air pre-heater.

Working Principle:

Step 1 – Fuel Combustion: Coal (or oil/gas) is fed into the furnace via coal feeders. Combustion produces flue gases at high temperature (1400–1600°C).

Step 2 – Steam Generation: Heat from flue gases is transferred to water in the boiler drum. Water converts into high-pressure steam (~150–200 bar, ~540°C).

Step 3 – Superheating: Steam passes through the superheater, further increasing its temperature to improve turbine efficiency and prevent condensation in turbine blades.

Step 4 – Turbine Work: High-pressure superheated steam expands through turbine stages (HP→IP→LP), rotating the turbine at 3000 RPM (for 50 Hz generation). The turbine shaft is coupled to the generator.

Step 5 – Power Generation: The alternator/generator converts mechanical energy to 3-phase AC electricity (11–22 kV), which is stepped up by transformer for grid transmission.

Step 6 – Condensation: Exhaust steam from LP turbine (low pressure) enters the condenser. Cooling water (from river/cooling tower) condenses steam back to water.

Step 7 – Cycle Completion: Condensate is pumped by the feed pump through the economiser (heated by flue gases) back to the boiler. The cycle repeats.

Heat Recovery: Hot flue gases leaving the furnace pass through superheater → reheater → economiser → air pre-heater → electrostatic precipitator (for ash) → chimney. This improves overall efficiency.

Efficiency: η = Net Electrical Output / Heat Input of Fuel. Typical efficiency = 30–40%.

Advantages: High capacity, reliable, suitable for base-load operation. **Disadvantages:** High pollution (CO_2, SO_2, NO_x), low efficiency, high water requirement.

5 Marks **Q2. Explain Belt Drive with a neat sketch. Derive the formula for velocity ratio with slip.**

Definition: A belt drive transmits power between two parallel shafts using a flexible belt running over pulleys by means of friction between belt and pulley surface.

Construction: Two pulleys (driver and driven) are mounted on parallel shafts. A flexible belt (flat, V, or ribbed) is looped around both pulleys under tension. The tight side carries more tension (T_1) while the slack side has less tension (T_2).

Working Principle: The motor/engine drives the driver pulley which rotates. The belt in contact with driver pulley rotates due to frictional force. This force in the belt rotates the driven pulley, which in turn drives the machine. The power transmitted = $(T_1 - T_2) \times v$, where v is belt velocity.

Velocity Ratio Derivation (with slip):

Let: D_1 = driver diameter, D_2 = driven diameter, N_1 = driver speed, N_2 = driven speed, s_1 = slip on driver side (%), s_2 = slip on driven side (%), s = total slip (%)

Belt velocity at driver: $v = \pi D_1 N_1 / 60 \times (1 - s_1/100)$

Belt velocity at driven: $v = \pi D_2 N_2 / 60 \times (1 + s_2/100)$

Equating: $N_2/N_1 = D_1(1-s_1/100) / D_2(1+s_2/100)$

For small slip: $N_2/N_1 \approx (D_1/D_2)(1 - s/100)$, where $s = s_1 + s_2$ = total % slip

Key Formulas:

- Ratio of tensions: $T_1/T_2 = e^{(\mu\theta)}$ where μ = friction coefficient, θ = angle of wrap (radians)
- Power: $P = (T_1 - T_2) \times v = (T_1 - T_2) \times \pi D_1 N_1/60$ [Watts]
- Centrifugal tension: $T_c = mv^2$ (reduces effective tension)
- Effective tension: $T_1 - T_c$ to $T_2 - T_c$ ratio equals $e^{(\mu\theta)}$

Applications: Lathe machines, textile mills, flour mills, fans, pumps, agricultural equipment.

Advantages: Smooth silent operation, shock-absorbing, low cost, overload protection. **Disadvantages:** Slip causes velocity inaccuracy, belt deteriorates with oil/heat, not suitable for exact timing applications.

5 Marks **Q3. Compare Steam, Diesel, Hydro, and Nuclear Power Plants on various parameters.**

Introduction: Power plants convert different forms of primary energy into electrical energy. Each type has distinct advantages, limitations, and suitable applications. The choice depends on geography, fuel availability, load requirement, and environmental considerations.

1. Steam (Thermal) Power Plant: Uses fossil fuels (coal, oil, gas). Works on Rankine cycle. Efficiency 30–40%. Suitable for large-scale base-load (100–2000 MW). High CO_2 emissions. Can be built anywhere. High running cost but proven technology. e.g., NTPC Vijayawada, Ramagundam.

2. Diesel Power Plant: Uses CI engine with diesel as fuel. Efficiency 35–45%. Small to medium capacity (1–50 MW). Quick start in minutes. Very high fuel cost; not suitable for base load. Used as standby/emergency power for hospitals, data centers, remote areas.

3. Hydro Power Plant: Uses potential energy of stored water. Efficiency 85–95% (highest). No fuel cost after installation; renewable. Very high capital cost; site-specific (needs river/dam). Long construction time. No air pollution. Used for base and peak load. e.g., Nagarjunasagar (816 MW), Srisailam (1670 MW).

4. Nuclear Power Plant: Uses fission of U-235. Efficiency ~33–40% (similar to thermal). Very high capacity (500–3000 MW) from small fuel mass. No CO_2 emissions but radioactive waste problem. Very high capital cost; complex safety requirements. e.g., Kudankulam (2000 MW), Tarapur.

Key Differences in Tabular Summary (as answers above): Hydro is the most efficient and renewable. Diesel is fastest to start but costliest to run. Nuclear has the highest energy density. Steam is the most widely used for base load in India.

Conclusion: The selection of power plant depends on availability of natural resources, load characteristics, economics, and environmental regulations. India uses a mix of all four types to meet its diverse power needs.

5 Marks Q4. Explain Robot Configurations with neat sketches and compare them.

Introduction: A robot configuration defines the geometric arrangement of joints and links that determines the shape of the workspace (work envelope) and the type of motion the robot can perform.

1. Cartesian Configuration (PPP – Three Prismatic Joints): Three linear slide axes (X, Y, Z) arranged mutually perpendicular. Workspace is a rectangular box. High positional accuracy; simple kinematics. Used in CNC machines, gantry robots, 3D printers, overhead cranes. Disadvantage: Large floor space; cannot reach around obstacles.

2. Cylindrical Configuration (RPP – 1 Revolute + 2 Prismatic): Rotary base + vertical slide + horizontal telescopic arm. Workspace is a hollow cylinder (annular ring). Good horizontal reach; suitable for loading/unloading operations. Used in machine loading/unloading, spot welding, assembly. Disadvantage: Cannot reach behind base; limited workspace variety.

3. Spherical (Polar) Configuration (RRP – 2 Revolute + 1 Prismatic): Two rotary joints + one prismatic (radial) arm. Workspace is a partial sphere. Large work volume relative to robot size. Used in die casting, forging, material handling, spray coating. Disadvantage: Complex kinematics; less accurate at far reaches.

4. Articulated (Revolute/Jointed-arm) Configuration (RRR – All Revolute): All three (or more) joints are revolute, mimicking the human arm (waist + shoulder + elbow + wrist). Most flexible; largest work envelope for given robot size. 6 DOF version can reach any position and orientation. Used for arc welding, spray painting, assembly, palletising. Most widely used in industry. Disadvantage: Complex kinematics; programming is harder.

5. SCARA Configuration (RRP – Selective Compliance Articulated Robot Arm): Two horizontal revolute joints + one vertical prismatic. Stiff in vertical direction but compliant horizontally. Very fast and precise for horizontal planar tasks. Used in PCB assembly, electronic component placement, pick & place operations.

Comparison: Articulated robots offer the most versatility and are used in 70%+ of industrial applications. Cartesian robots are simplest to program. SCARA robots are fastest for planar tasks. Spherical/cylindrical are used for medium-complexity tasks.

Conclusion: The choice of robot configuration depends on the workspace shape required, payload, speed, accuracy, and the nature of the industrial task.

5 Marks Q5. Explain Gear Drives: types, advantages, and formulas.

Definition: A gear drive transmits power and motion between shafts through interlocking toothed wheels (gears). It is a positive (no-slip) drive providing an exact, constant velocity ratio.

Why Gears? Gears are used when: (1) exact speed ratio is required, (2) high torque multiplication is needed, (3) shafts are at different orientations (parallel, intersecting, or skew).

Types of Gears:

Spur Gear: Teeth parallel to gear axis; used for parallel shafts; noisy at high speed. Used in clocks, simple gearboxes, bicycles.

Helical Gear: Teeth inclined at helix angle; smooth and quiet; can carry more load than spur gear. Used in automotive transmissions, turbine reduction boxes.

Bevel Gear: Conical gears for intersecting shafts (usually 90°). Straight or spiral bevel. Used in automotive differential, bevel gearboxes, drill machines.

Worm & Worm Gear: Worm (screw) meshes with worm wheel; non-parallel, non-intersecting shafts; very high reduction ratio (up to 100:1); self-locking property. Used in lifts, reduction gearboxes, steering.

Rack & Pinion: Converts rotary to linear motion. Used in steering systems, CNC machines, elevators.

Key Formulas:

- Gear ratio: $i = T_2/T_1 = N_1/N_2$ (T = number of teeth)
- Module: $m = D/T$ (D = pitch circle diameter in mm, T = no. of teeth)
- Pitch: $p = \pi m$ (circular pitch)
- Centre distance: $C = m(T_1+T_2)/2$
- Gear train velocity ratio: TV = N_last/N_first = Product of driver teeth / Product of driven teeth

Advantages: No slip (exact ratio); very high efficiency (96–99%); compact; high power capacity; long service life; multiple speed ratios in a gearbox. **Disadvantages:** High cost of manufacture; require lubrication; noise (spur gears); no overload slip protection; vibration from tooth profile errors.

Applications: Automotive gearboxes; machine tool headstocks; wind turbine gearboxes; watch mechanisms; crane hoists; industrial reducers; all precision mechanisms requiring exact speed control.

SECTION 16

⚡ Quick Revision Sheet – Unit III

🔥 Power Plants – Summary

- **Steam:** Coal → Boiler → Steam → Turbine → Generator; $\eta \approx 30$–40%
- **Diesel:** Fuel injection → CI Engine → Crankshaft → Generator; $\eta \approx 35$–45%
- **Hydro:** Dam → Penstock → Water Turbine → Generator; $\eta \approx 85$–95%
- **Nuclear:** U-235 fission → Heat → Steam → Turbine → Generator; $\eta \approx 33$–40%

📐 Key Power Plant Formulas

- $\eta = W\_net / Q\_in \times 100\%$
- P (hydro) = $\rho gQH \times \eta$
- $E = mc^2$ (nuclear)
- IP = $P_m \cdot L \cdot A \cdot N/60$ (diesel)
- BP = $2\pi NT/60$ [Watts]
- Heat Rate = $3600/\eta$ [kJ/kWh]

⚙ Belt Drive – Key Points

- Types: Flat, V-belt, Ribbed, Timing, Round
- Open belt: same direction; crossed: opposite
- $VR = N_2/N_1 = D_1/D_2$ (no slip)
- VR with slip = $D_1/D_2 \times (1 - s/100)$
- $T_1/T_2 = e^{(\mu\theta)}$ — belt tension ratio
- $P = (T_1 - T_2) \times v$ [Watts]
- Slip: 1–2%; reduces speed ratio

⛓ Chain & Rope Drives

- **Chain:** Positive drive; no slip; $VR = T_2/T_1$
- Types: Roller, Silent, Leaf, Flat-link
- Chain speed = $T \cdot p \cdot N/60$ [m/s]
- **Rope:** Friction drive over grooved sheaves
- Types: Fibre (hemp), Wire rope (6×7, 6×19, 6×37)
- Wire rope: used in cranes, elevators, mines

⚙ Gear Drive – Key Points

- Types: Spur, Helical, Bevel, Worm, Rack & Pinion
- Gear ratio = $T_2/T_1 = N_1/N_2$
- Module $m = D/T$
- Pitch $p = \pi m$
- Centre dist $C = m(T_1+T_2)/2$
- η gear ≈ 96–99% (highest efficiency)
- Worm: self-locking, high reduction (up to 100:1)

🤖 Robotics – Key Points

- **Joints:** Revolute (R), Prismatic (P), Cylindrical (C), Spherical (S), Screw (H)
- **Configs:** Cartesian (PPP), Cylindrical (RPP), Spherical (RRP), Articulated (RRR), SCARA (RRP), Delta
- DOF = sum of joint DOF
- 6 DOF for full position + orientation
- Articulated = most common industrial robot
- SCARA = fastest for planar PCB assembly

Power Plant Components – At a Glance

Power Plant	Prime Mover	Fuel/Medium	Unique Component	Efficiency
Steam	Steam Turbine	Coal/Oil/Gas	Superheater, Economiser	30–40%
Diesel	CI Engine	Diesel	Fuel Injector, Flywheel	35–45%
Hydro	Water Turbine	Water (head H)	Penstock, Surge Tank	85–95%
Nuclear	Steam Turbine	U-235	Reactor, Control Rods, Biological Shield	33–40%

Drive System – Comparison Quick Reference

Feature	Belt	Chain	Rope	Gear
Slip?	Yes	No	Yes	No
Efficiency	95–98%	97–99%	90–95%	96–99%
Distance	Long	Short-Med	Very Long	Very Short
Cost	Low	Medium	Medium	High
Overload Protection	Yes (slip)	No	Yes	No

Robotics Quick Reference

Configuration	Joints	Workspace	Best For
Cartesian	PPP	Box	CNC, 3D Printing
Cylindrical	RPP	Cylinder	Loading/Unloading
Spherical	RRP	Sphere	Die Casting, Forging
Articulated	RRR	Irregular	Welding, Painting, Assembly
SCARA	RRP	Flat Cylinder	PCB Assembly, Pick & Place
Delta (Parallel)	RRR (parallel)	Hemisphere	Food/Pharma Packaging

SECTION 17

📚 Textbook & Reference Books

Prescribed Textbooks

📖 **V. Ganesan** – *Internal Combustion Engines*, 4th Edition, Tata McGraw-Hill, New Delhi. [Diesel Engine, Combustion, Thermodynamic Cycles]

📖 **S.S. Rattan** – *Theory of Machines*, S. Chand & Company, New Delhi. [Belt Drives, Chain Drives, Gear Drives, Speed Ratio, Kinematics of Mechanisms]

📖 **Jonathan Wickert & Kemper Lewis** – *An Introduction to Mechanical Engineering*, 4th Edition, Cengage Learning. [Power Transmission, Fluid Power, Introduction to Engineering Design]

Reference Books

📖 **G. Shanmugam & M.S. Palanisamy** – *Basic Civil & Mechanical Engineering*, Anuradha Publishers. [As per JNTUK Syllabus – comprehensive treatment of all Unit III topics]

📖 **Mahesh M. Rathore** – *Thermal Engineering*, McGraw-Hill India. [Steam Power Plants, Diesel Cycles, Nuclear Energy, Thermodynamic Analysis]

📖 **L. Jyothish Kumar & Pulak M. Pandey** – *3D Printing & Additive Manufacturing: Principles & Applications*, Universities Press. [Advanced Manufacturing, Additive Processes – supplementary for modern engineering context]

📖 **Appu Kuttan K.K.** – *Robotics*, Prentice Hall of India. [Robot Kinematics, Robot Configurations, End Effectors, Sensors, Programming, Industrial Applications – primary reference for Robotics portion]

Online Resources (Supplementary)

📖 **NPTEL** – National Programme on Technology Enhanced Learning: *Basic Mechanical Engineering* (IIT Kharagpur), *Power Plant Engineering* (IIT Roorkee)

📖 **SWAYAM Portal** – MOOCs by Government of India for Engineering courses aligned with AICTE/UGC syllabus

📖 **Virtual Labs (IIT/NIT)** – Online simulation of power plant cycles, gear systems, and robot simulations

Important Journals & Standards

Standard / Journal	Relevance to Unit III
IS 2091 (BIS)	V-belt specifications and dimensions (India)
ANSI/ASME B29.1	Roller chain dimensions and tolerances

Standard / Journal	Relevance to Unit III
ISO 8373:2012	Robots & robotic devices – vocabulary and definitions
IEEE Transactions on Power Systems	Power plant operation, grid integration, nuclear safety
ASME Journal of Mechanical Design	Gear design, power transmission research

📌 Exam Tips – Unit III

- Always **draw diagrams** for power plants and drive systems — fetch 2–3 marks easily
- Memorise the **energy conversion chain** for each power plant type
- For belt drives: remember both the **velocity ratio formula WITH and WITHOUT slip**
- For gears: know the **module formula** (m = D/T) and gear ratio formula
- For robotics: remember **joint symbols (R, P)** and workspace shapes for each configuration
- Comparison tables between power plants and drive systems are **frequently asked for 5 marks**
- Nuclear power plant: explain **chain reaction, control rods, and biological shielding** — guaranteed marks
- Hydro power: mention **surge tank function (water hammer)** for extra marks